

PRODUCT AND CATEGORY MANAGEMENT

StyleNova – Fashion Products E-Commerce Order Management System

1. Database Design

1.1 Category Table

Attribute	Data Type	Constraint	Description
Category_ID	INT	Primary Key	Unique identifier for each category
Category_Name	VARCHAR(100)	NOT NULL, UNIQUE	Name of the product category
Description	VARCHAR(200)	—	Description of the category

1.2 Product Table

Attribute	Data Type	Constraint	Description
Product_ID	INT	Primary Key	Unique identifier for each product
Product_Name	VARCHAR(100)	NOT NULL	Name of the fashion product
Category_ID	INT	Foreign Key	References Category.Category_ID
Price	NUMBER(10,2)	NOT NULL	Price of the product
Stock_Quantity	INT	NOT NULL	Available quantity of the product

2. Primary and Foreign Key Relationship

Category → Product

- Category.Category_ID → Primary Key
- Product.Product_ID → Primary Key
- Product.Category_ID → Foreign Key referencing Category.Category_ID
- Relationship: One-to-Many (1:M)

3. Table Creation

3.1 Create Category Table

```
SQL> CREATE TABLE StyleNova_Category (  
2     Category_ID NUMBER PRIMARY KEY,  
3     Category_Name VARCHAR2(100) NOT NULL UNIQUE,  
4     Description VARCHAR2(200)  
5 );
```

Table created.

3.2 Create Product Table

```
SQL> CREATE TABLE StyleNova_Product (  
 2     Product_ID NUMBER PRIMARY KEY,  
 3     Product_Name VARCHAR2(100) NOT NULL,  
 4     Category_ID NUMBER NOT NULL,  
 5     Price NUMBER(10,2) NOT NULL,  
 6     Stock_Quantity NUMBER NOT NULL,  
 7     CONSTRAINT fk_stylenova_category  
 8         FOREIGN KEY (Category_ID)  
 9         REFERENCES StyleNova_Category(Category_ID)  
10 );
```

Table created.

4. Data Insertion

4.1 Category Records

```
SQL> INSERT INTO StyleNova_Category  
 2 (Category_ID, Category_Name, Description)  
 3 VALUES (1, 'Men', 'Fashion products for men');
```

1 row created.

```
SQL>  
SQL> INSERT INTO StyleNova_Category  
 2 (Category_ID, Category_Name, Description)  
 3 VALUES (2, 'Women', 'Fashion products for women');
```

1 row created.

```
SQL>  
SQL> INSERT INTO StyleNova_Category  
 2 (Category_ID, Category_Name, Description)  
 3 VALUES (3, 'Kids', 'Fashion products for kids');
```

1 row created.

```
SQL>  
SQL> INSERT INTO StyleNova_Category  
 2 (Category_ID, Category_Name, Description)  
 3 VALUES (4, 'Accessories', 'Fashion accessories');
```

1 row created.

4.2 Product Records

```
SQL> INSERT INTO StyleNova_Product
  2 (Product_ID, Product_Name, Category_ID, Price, Stock_Quantity)
  3 VALUES (102, 'Denim Jeans', 1, 1299.00, 15);

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Product
  2 (Product_ID, Product_Name, Category_ID, Price, Stock_Quantity)
  3 VALUES (103, 'Floral Kurti', 2, 899.00, 20);

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Product
  2 (Product_ID, Product_Name, Category_ID, Price, Stock_Quantity)
  3 VALUES (104, 'Kids Hoodie', 3, 749.00, 12);

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Product
  2 (Product_ID, Product_Name, Category_ID, Price, Stock_Quantity)
  3 VALUES (105, 'Leather Handbag', 4, 1499.00, 10);

1 row created.

SQL>
SQL> INSERT INTO StyleNova_Product
  2 (Product_ID, Product_Name, Category_ID, Price, Stock_Quantity)
  3 VALUES (106, 'Women Sneakers', 2, 1199.00, 18);

1 row created.

SQL>
SQL> COMMIT;

Commit complete.
```

5. Data Retrieval

5.1 Display Category Records

```
SQL> SELECT * FROM StyleNova_Category;
```

CATEGORY_ID	CATEGORY_NAME	DESCRIPTION
1	Men	Fashion products for men
2	Women	Fashion products for women
3	Kids	Fashion products for kids
4	Accessories	Fashion accessories

5.2 Display Product Records

```
SQL> SELECT * FROM StyleNova_Product;

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE STOCK_QUANTITY
-----
101
Cotton T-Shirt
1             599           25

102
Denim Jeans
1            1299           15

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE STOCK_QUANTITY
-----
103
Floral Kurti
2             899           20

104
Kids Hoodie

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE STOCK_QUANTITY
-----
3             749           12

105
Leather Handbag
4            1499           10

106

PRODUCT_ID
-----
PRODUCT_NAME
-----
CATEGORY_ID    PRICE STOCK_QUANTITY
-----
Women Sneakers
2            1199           18

6 rows selected.

SQL> |
```

6. Product Update

Update the price and stock quantity of Cotton T-Shirt.

```
SQL> UPDATE StyleNova_Product
2 SET Price = 649.00,
3     Stock_Quantity = 30
4 WHERE Product_Name = 'Cotton T-Shirt';

1 row updated.
```

7. Product Deletion

Delete the product Kids Hoodie.

```
SQL> DELETE FROM StyleNova_Product
2 WHERE Product_Name = 'Kids Hoodie';

1 row deleted.
```

8. Category-wise Product Reports

8.1 Product Listing by Category

This report displays the category name along with the products belonging to each category.

```
SQL> SELECT c.Category_Name,  
2         p.Product_ID,  
3         p.Product_Name,  
4         p.Price,  
5         p.Stock_Quantity  
6 FROM StyleNova_Category c  
7 JOIN StyleNova_Product p  
8 ON c.Category_ID = p.Category_ID  
9 ORDER BY c.Category_Name, p.Product_Name;
```

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
Accessories	105	
Leather Handbag	1499	10

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
Men	101	
Cotton T-Shirt	649	30

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
Men	102	
Denim Jeans	1299	15

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
Women	103	
Floral Kurti	899	20

CATEGORY_NAME

PRODUCT_ID

PRODUCT_NAME

	PRICE	STOCK_QUANTITY
Women	106	
Women Sneakers	1199	18

8.2 Products in Women Category

```
SQL> SELECT p.Product_ID,  
2         p.Product_Name,  
3         p.Price,  
4         p.Stock_Quantity  
5 FROM StyleNova_Product p  
6 JOIN StyleNova_Category c  
7 ON p.Category_ID = c.Category_ID  
8 WHERE c.Category_Name = 'Women';
```

PRODUCT_ID

PRODUCT_NAME

PRICE STOCK_QUANTITY

103	
Floral Kurti	
899	20

106	
Women Sneakers	
1199	18

PRODUCT_ID

PRODUCT_NAME

PRICE STOCK_QUANTITY

8.3 Product Count by Category

```
SQL> SELECT c.Category_Name,  
2         COUNT(p.Product_ID) AS Product_Count  
3 FROM StyleNova_Category c  
4 LEFT JOIN StyleNova_Product p  
5 ON c.Category_ID = p.Category_ID  
6 GROUP BY c.Category_ID, c.Category_Name  
7 ORDER BY c.Category_Name;
```

CATEGORY_NAME

PRODUCT_COUNT

Accessories	1
-------------	---

Kids	0
------	---

Men	2
-----	---

CATEGORY_NAME

PRODUCT_COUNT

Women	2
-------	---